# Radio over Fiber (RoF)

Technology, Architecture, Advantages, Applications and Limitations

# Agenda

1. What is Radio over Fiber?
2. How it works: architecture and building blocks
3. Analog vs. digital RoF
4. Advantages of RoF
5. RoF in Distributed Antenna Systems (DAS)
6. Application areas
7. Limitations, challenges and outlook

# Why Do We Need Radio over Fiber?

## How do you bring a clear mobile signal inside a tunnel or a big stadium?

- **The problem:** walls, distance and crowds weaken radio signals, so one antenna cannot cover a whole building or venue.
- **The usual fix:** place many small antennas around the area, all fed from one central equipment room.
- **Coaxial cable** can feed them, but at radio frequencies it loses signal quickly over distance and is heavy and bulky.
- **Optical fiber** loses very little signal, is thin and light, and does not pick up electromagnetic interference. This is the idea behind RoF.

# RoF vs. Coaxial Cable and Microwave Links

| Aspect | Coaxial cable | Microwave link | RoF (fiber) |
|---|---|---|---|
| **Loss over distance** | High at radio frequencies; grows with frequency and length | Free-space path loss; needs a clear line of sight | Very low (about 0.2 dB/km at 1550 nm) |
| **Reach** | Short runs | Line-of-sight hops between towers | Many kilometers |
| **Interference** | Shielded, but not immune | Affected by weather and radio interference | Fiber span is immune to electromagnetic pickup |
| **Size and weight** | Heavy and bulky | Needs antennas, dishes and towers | Thin and light |
| **Bandwidth** | Moderate; loss limits useful length | Limited by allocated spectrum | Very large (more with WDM) |
| **Best fit** | Short indoor links | Point-to-point links where fiber is not available | Longer links to many remote antennas |

Qualitative comparison; actual values depend on cable type, frequency and link design. Fiber loss: ITU-T G.652 family.

# Example: Signal Loss over Distance

A 2.4 GHz signal carried over each medium (loss = power lost along the cable):

| Distance | Coax (about 22 dB per 100 m at 2.4 GHz) | Fiber (0.2 dB/km at 1550 nm) |
|---|---|---|
| **100 m** | about 22 dB loss (under 1% of the power left) | 0.02 dB loss (about 99.5% left) |
| **1 km** | about 220 dB loss | 0.2 dB loss |
| **10 km** | about 2,200 dB loss: not usable | 2 dB loss (about 63% left) |

**Takeaway:** over 10 km, fiber loses a few dB while coax loses far more than any amplifier could recover.

In an intensity-modulated link, 1 dB of optical loss reduces the recovered RF power by about 2 dB, so 2 dB of fiber loss is roughly 4 dB of RF loss. Still tiny next to coax.

Illustrative values. Coax: typical datasheet figure for LMR-400-class cable at 2.4 GHz; check your own cable's datasheet. Fiber: ITU-T G.652 family.

# Radio over Fiber: Definition & Principle of Operation

• **Core concept:** RoF transports an analog RF or IF waveform directly over an optical fiber carrier, rather than baseband digital bits.

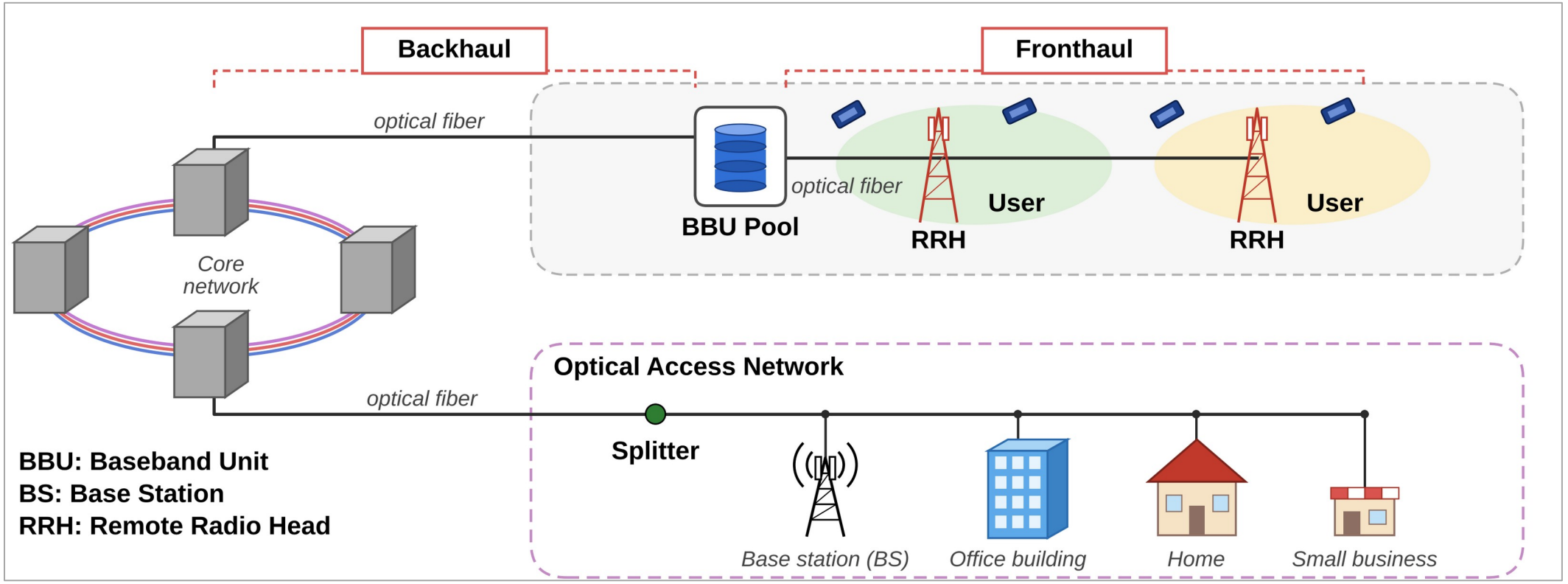
• **Downlink (E/O → O/E):** at the central station, the RF/IF signal modulates a CW laser carrier (direct modulation or an external MZM). At the remote unit, a photodiode recovers the RF/IF signal for filtering, amplification and antenna transmission.

• **Uplink:** follows the reverse conversion path, from the remote antenna back to the central processing unit.

• **Multiplexing:** wavelength-division multiplexing (WDM) lets several RF channels share one fiber on different wavelengths (λ1 … λn).

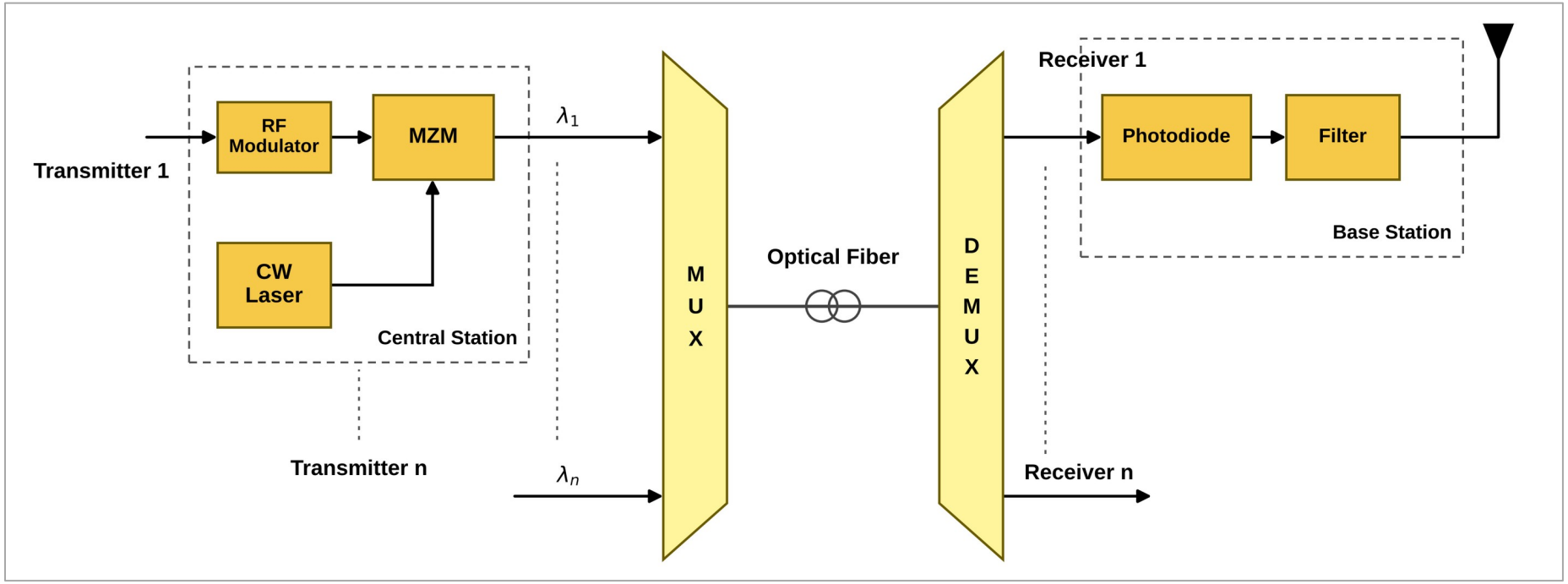
**Critical distinction:** *RoF removes the transport bottleneck, but it does not create extra radio spectrum or wireless air-interface capacity by itself.*

# RoF Network Architecture




BBU: Baseband Unit · BS: Base Station · RRH: Remote Radio Head

# RoF Link Block Diagram




**CW laser:** continuous-wave source that provides the optical carrier.

**MZM:** Mach-Zehnder modulator; varies light intensity with the RF signal (alternative: direct laser modulation).

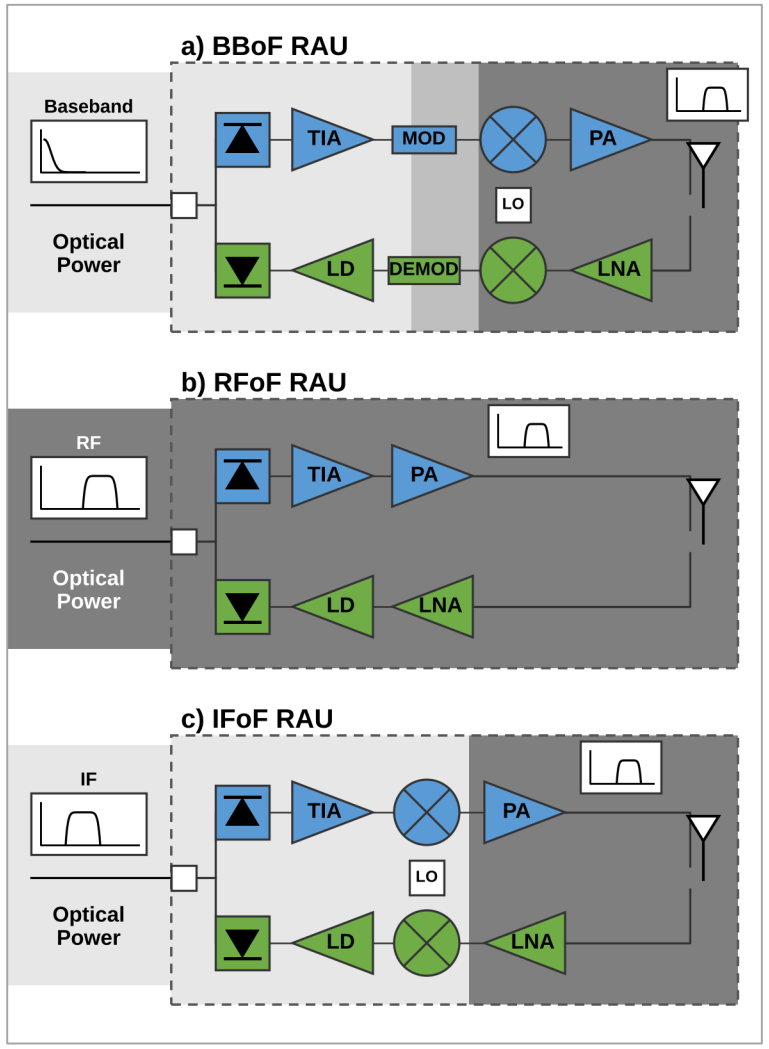
**WDM MUX / DEMUX:** combine and separate channels λ1 … λn when they share one fiber.

**Photodiode:** converts light back into the electrical RF signal at the base station (remote antenna site).

**Filter and PA:** remove out-of-band products and amplify the RF signal before the antenna.

**Downlink shown;** the uplink follows the reverse conversion path.

# RoF Remote Antenna Unit (RAU) Types




**BBoF — baseband over fiber**

RAU: modulator + mixer + LO. Converts baseband up to RF.

**IFoF — intermediate frequency over fiber**

RAU: mixer + LO. Converts IF up to RF.

**RFoF — RF over fiber**

RAU: photodiode + TIA + PA. No frequency conversion; simplest RAU.

**All types**

Downlink: photodiode, TIA, PA.  Uplink: LNA, laser diode (LD).

TIA: transimpedance amplifier · PA: power amplifier · LNA: low-noise amplifier. Figure adapted from “Radio over Fiber: An Alternative Broadband Network Technology for IoT,” Electronics 9(11), 1785 (2020), Fig. 3.

# Analog vs. Digital RoF

| Aspect | Analog RoF (A-RoF) | Digital RoF (D-RoF) |
|---|---|---|
| Signal on fiber | The RF waveform itself modulates the light | Sampled and quantized digital bits |
| Remote unit | Very simple: photodiode, filter, amplifier | More complex: needs DAC / ADC |
| Fiber bandwidth | Efficient | Very high; grows with antennas and bandwidth |
| Main weakness | Noise, nonlinearity, dispersion | High data rate, remote-unit cost and power |
| Standards | ITU-T G.9803 defines RoF system requirements, including IMT-2020 fronthaul in Amendment 2 (2022). | CPRI / eCPRI: digital fronthaul interface families. |

Sources: Li et al. (2021); ITU-T G.9803 (2018), Amendment 2 (2022).

# Key Engineering Advantages of RoF

**1. Ultra-low attenuation & extended reach**

≈ 0.2 dB/km at 1550 nm (ITU-T G.652 SMF), so one central site can serve a wide coverage area. Reach is bounded by noise, nonlinearity and dispersion.

**2. Large transport bandwidth**

High fiber capacity, extended with WDM, supports multiple channels and operators on one fiber.

**3. EMI immunity**

The optical fiber span is immune to electromagnetic interference. Over-the-air RF interference still applies.

**4. Centralized processing, low field maintenance**

Complex processing stays at the BBU pool. Simpler remote units with fewer active parts reduce remote power and maintenance.

**5. Protocol & modulation transparency**

A transparent optical link carries multiple radio generations and modulation formats without changing remote hardware.

**6. Cost & installation efficiency**

Light, flexible cable simplifies installation in tunnels and buildings; existing dark fiber can be reused.

**Note:** wireless capacity remains governed by radio spectrum, modulation efficiency and the number of radio units, not by fiber bandwidth alone.

Fiber attenuation: typical single-mode fiber at 1550 nm, ITU-T G.652 family.

# RoF: Distributed Antenna System (DAS)

- A common RoF deployment model is the Distributed Antenna System (DAS), which connects a central equipment room to remote radio units over fiber.
- DAS can improve indoor or venue coverage in places such as stadiums, airports, shopping malls and campuses. Wireless capacity still depends on radio spectrum and radio design.